



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Properties of the Riemann Integral

**Algebraic Properties, Order Properties, Additivity Property and
Uniform Convergence in Riemann Integral**

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CONDUCTED BY

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Cauchy Criterion for Riemann Integrability

Riemann integrable function

A function f is said to be Riemann integrable on the interval $[a, b]$ if for every $\epsilon > 0$, there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \epsilon,$$

where $U(f, P)$ and $L(f, P)$ are the upper and lower sums of f with respect to the partition P , respectively.

Algebraic Properties of the Riemann Integral

💡 Scalar multiplication

Let f be Riemann integrable functions on the interval $[a, b]$ and $k \in \mathbb{R}$. Then, kf is Riemann integrable on $[a, b]$, and

$$\int_a^b kf = k \int_a^b f.$$

💡 Sum of integrable functions

Let f and g be Riemann integrable functions on the interval $[a, b]$.
Then, $f + g$ is Riemann integrable on $[a, b]$, and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

💡 Other algebraic properties

Let f and g be Riemann integrable functions on the interval $[a, b]$. Then, the following properties hold:

1. $f - g$ is Riemann integrable on $[a, b]$, and

$$\int_a^b (f - g) = \int_a^b f - \int_a^b g.$$

2. fg is Riemann integrable on $[a, b]$.
3. If $f(x) \neq 0$ for all $x \in [a, b]$, then $\frac{1}{f}$ is Riemann integrable on $[a, b]$.
4. If $g(x) \neq 0$ for all $x \in [a, b]$, then $\frac{f}{g}$ is Riemann integrable on $[a, b]$.

Order Properties of the Riemann Integral

💡 Non-negativity

Let f be a Riemann integrable function on the interval $[a, b]$. If $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f \geq 0.$$

Order preservation

Let f and g be Riemann integrable functions on the interval $[a, b]$. If $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f \leq \int_a^b g.$$

Absolute value

Let f be a Riemann integrable function on the interval $[a, b]$. Then, $|f|$ is Riemann integrable on $[a, b]$, and

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Additivity of the Riemann Integral

Additivity

Let f be a Riemann integrable function on the interval $[a, b]$ and $c \in (a, b)$. Then,

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Reversal of the limits of integration

Integral from b to a

For a Riemann integrable function f on the interval $[a, b]$, the integral of f from b to a is defined as

$$\int_b^a f = - \int_a^b f.$$

Integral over a point

For a Riemann integrable function f on the interval $[a, b]$, the integral of f over a point is defined as

$$\int_a^a f = 0.$$

Uniform Convergence and the Riemann Integral

💡 Uniform convergence and the Riemann integral

Let $\{f_n\}$ be a sequence of Riemann integrable functions on the interval $[a, b]$ that converges uniformly to a function f . Then, f is Riemann integrable on $[a, b]$, and

$$\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f.$$

In other words,

$$\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b \lim_{n \rightarrow \infty} f_n.$$

Thank You!

We'd love your questions and feedback.

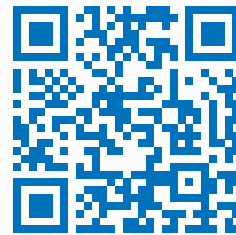
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(Lectures, walkthroughs, and course updates)



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References

- [1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.
- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.